# Spark! Project: Boston Bus Equity

CS506 October Check-In Report | Caleb Lee, Emir Kavukcu, Travis Meyer, Toriana Mullins, Ziad Alharbi

**Summary:** We use MBTA bus data to measure how service varies by route and area, and who poor service affects most.

### Key terms

| Term | What it means |
|---|---|
| **Timepoint** | A bus stop where the MBTA checks whether a bus is keeping to its schedule. A route has only a few of these. |
| **Crossing** | One bus passing one timepoint. A single trip creates several crossings. |
| **On time** | A crossing that passes the MBTA's reliability test. The on-time rate is the share of crossings that pass. |
| **Target routes** | 15 priority routes from the Livable Streets report: 14, 15, 17, 22, 23, 24, 26, 28, 29, 31, 33, 42, 44, 45, 111. |

## Data Processing

### What are our data sources, and how did we collect them?

All of our data is public, and our notebook downloads it automatically.

| Data | Where it came from | How we got it | What we used it for |
|---|---|---|---|
| **Bus arrival and departure times, 2020, 2024** | MBTA | Downloaded files | Reliability, wait, trip time |
| **Bus ridership by route, Fall 2016-2025** | MBTA | Downloaded files | Number of riders on each route |
| **Rider survey, 2025 (latest public)** | MBTA | API | Riders' race and income |
| **Rider survey, 2015-17** | Planning agency (CTPS) | Downloaded file | Comparison with earlier riders |
| **Bus stop locations** | MBTA | Downloaded file | Placing each stop on the map |
| **Census income and race, 2020-24** | Census Reporter | API | Who lives near each stop |
| **Census tract boundaries** | US Census | API | Matching stops to tracts |
| **Service Delivery Policy, Fall 2024** | MBTA | Read the policy | The definition of on time |

### What cleaning and processing did we do, and why?

| What we did | Why |
|---|---|
| **Compared Jan-Feb 2020 with Jan-Feb 2024** | These are the last pre-COVID months available, and using the same months removes seasonal effects. |
| **Kept weekdays only, and stopped 2024 at November** | Weekend schedules differ, and the network changed in December 2024. |
| **Matched route names across files (SL1 = 741)** | Files name routes differently, so busy routes would otherwise be lost. |
| **Left out 1.7 million blank arrival times** | A blank can be a canceled trip or a data gap. Counting them all as not on time lowers the 2024 on-time rate from 66.3% to 61.4%. |
| **Removed 2,025 extreme delays (over 30 min early or 2 hours late)** | These are likely tracking errors. |
| **Matched bus stops to census tracts** | This links bus performance to neighborhood income and race. |

**Check:** our Fall 2018 ridership is within 14% of the Livable Streets figures on all 15 target routes.

### Why did we choose these features?

| Feature | Why |
|---|---|
| **On-time rate** | This is the MBTA's own reliability standard. |
| **Average delay** | It shows how late buses run, which the on-time test can miss on frequent routes. |
| **Expected wait** | It estimates how long riders wait at the stop. |
| **Trip time** | It shows how long trips take and how predictable they are. |
| **Weekday boardings** | They show how many people each route affects. |
| **Race and income** | They show who is affected, both for riders (survey) and for neighborhoods (census). |

## Hypotheses Tested

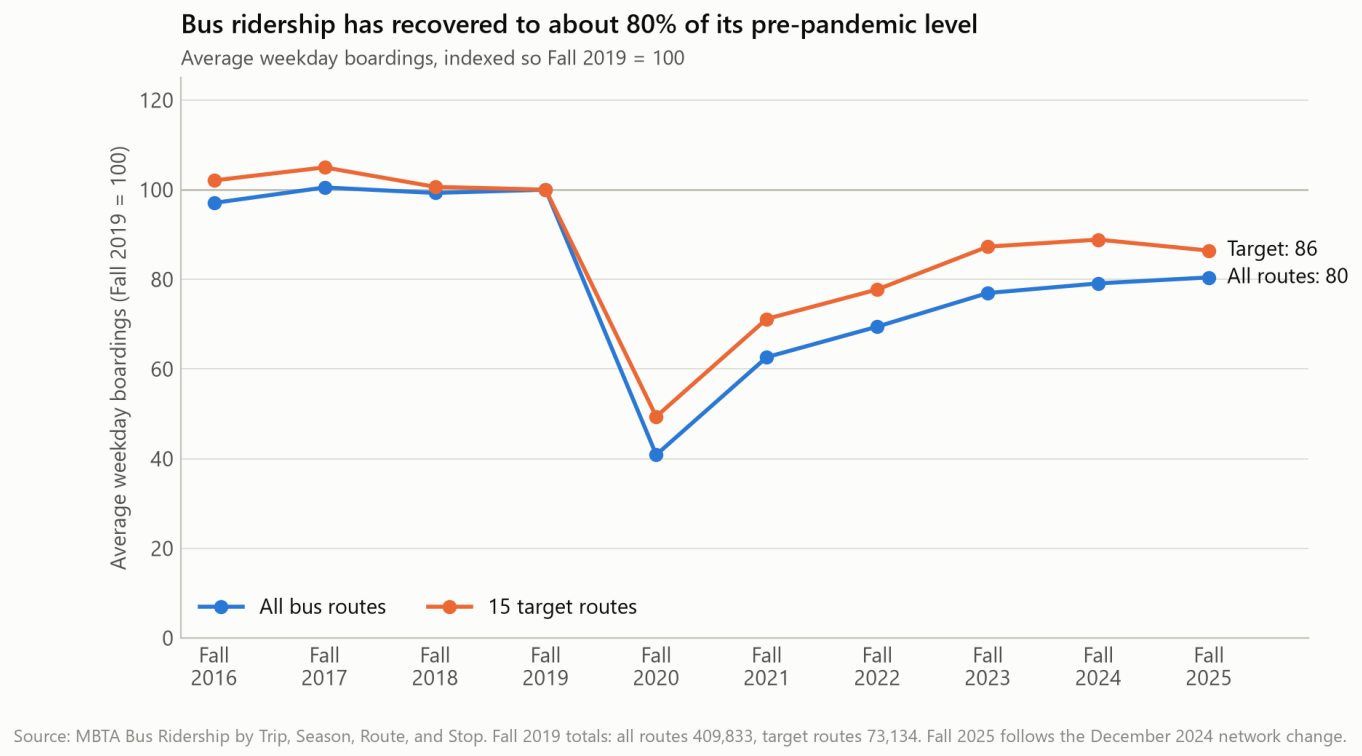
"Yes" means the data back up the claim, but it does not prove what caused it.

| # | What we expected | True? | What we found | Figures |
|---|---|---|---|---|
| **H1** | Fewer people ride the bus than before the pandemic, and the drop differs by route | **Yes** | Weekday riders are at 79% of their Fall 2019 level, and the typical route lost 20%. | 1, 2 |
| **H2** | Bus trips take longer and are less predictable than scheduled | **Partly** | A typical trip takes about as long as scheduled, but fast and slow trips on a route differ by about 16 minutes. | 3 |
| **H3** | Riders wait longer than the schedule suggests | **Yes** | On frequent routes, the wait is 8.6 minutes, not the 6.9 the schedule suggests. | 4 |
| **H4** | Some routes are much more reliable than others | **Yes** | Routes range from 40% to 80% on time. | 5 |
| **H5** | The 15 target routes perform worse than other routes | **Partly** | They are on time about as often, but run about 1.4 minutes later on average. | 5 |
| **H6** | Minority and low-income communities get worse service | **Partly** | On-time rates are about the same across neighborhoods, but buses run somewhat later in minority and lower-income areas. | 7, 8, 9 |
| **H7** | A small number of routes cause most of the lost time | **Weakly** | The 15 routes losing the most time account for 45% of lost hours, but they also carry 38% of riders. | 10 |
| **H8** | Buses were less reliable in 2024 than before the pandemic | **Yes** | From early 2020 to early 2024, the on-time rate fell from 73% to 69%, and average delay rose from 2.3 to 3.9 minutes. | 6 |

## Data Visualizations

### Figure 1. Ridership over time

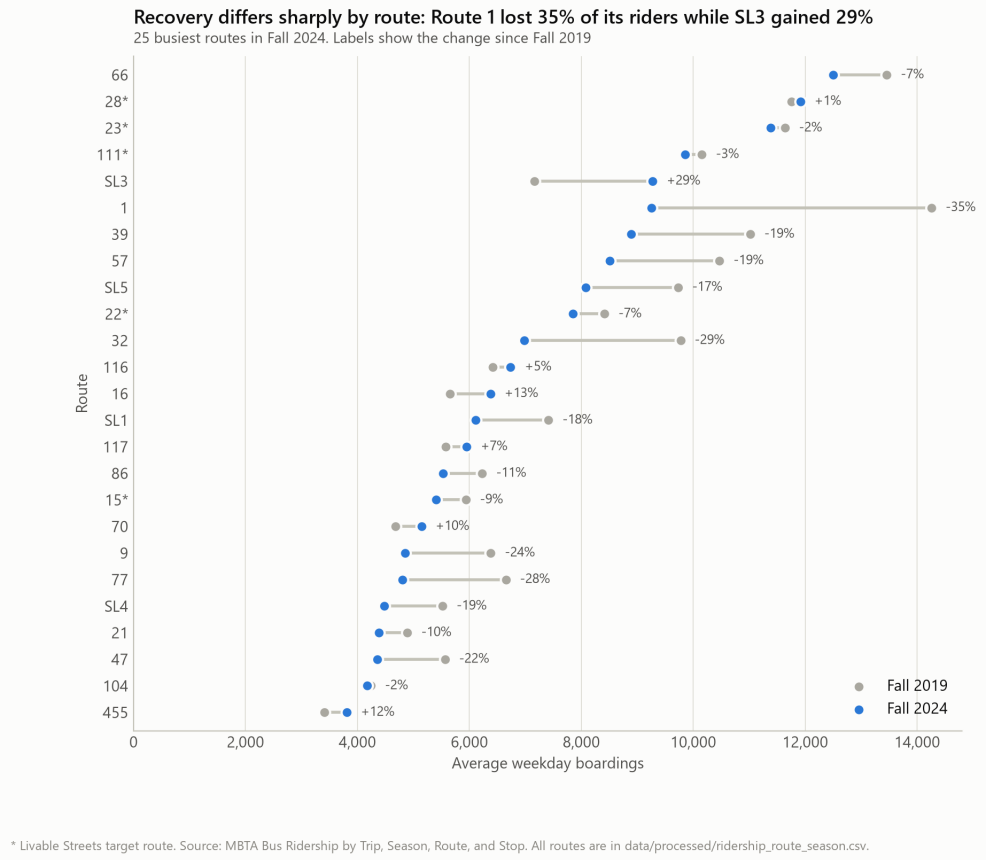
**Verdict:** True: bus ridership has not fully come back, with weekday riders at about 80% of their Fall 2019 level.




Source: MBTA Bus Ridership by Trip, Season, Route, and Stop. Fall 2019 totals: all routes 409,833, target routes 73,134. Fall 2025 follows the December 2024 network change.

## Figure 2. Ridership by route

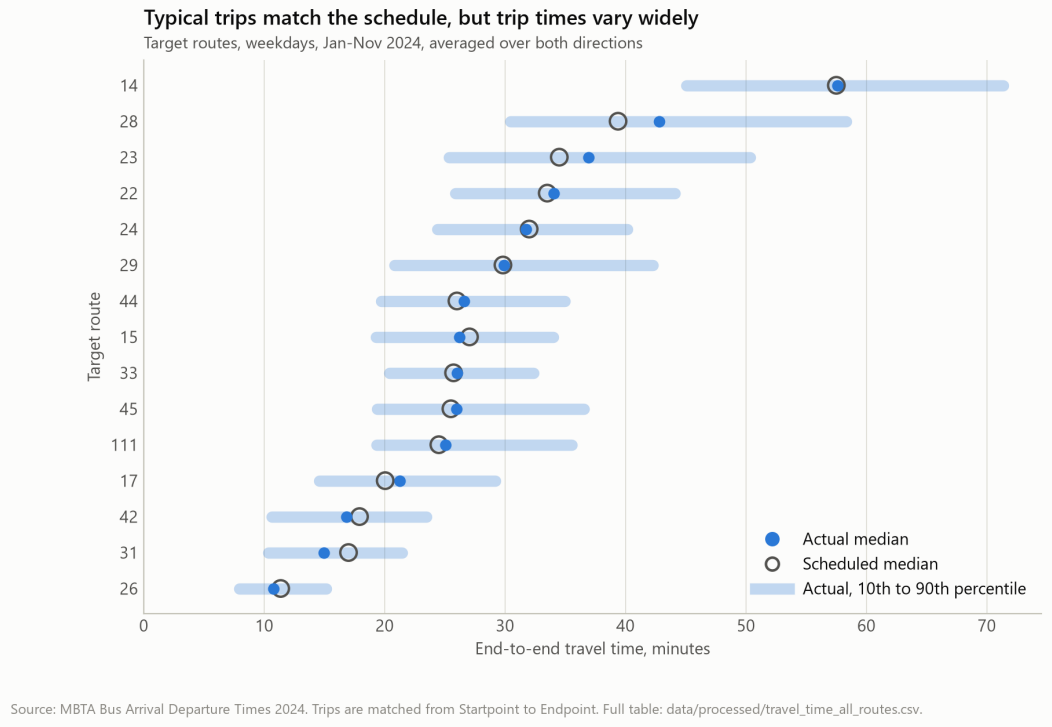
**Verdict:** True: the recovery is uneven, since the typical route lost 20% of its riders and only 18 of 127 routes gained riders.




* Livable Streets target route. Source: MBTA Bus Ridership by Trip, Season, Route, and Stop. All routes are in data/processed/ridership_route_season.csv.

## Figure 3. End-to-end travel time

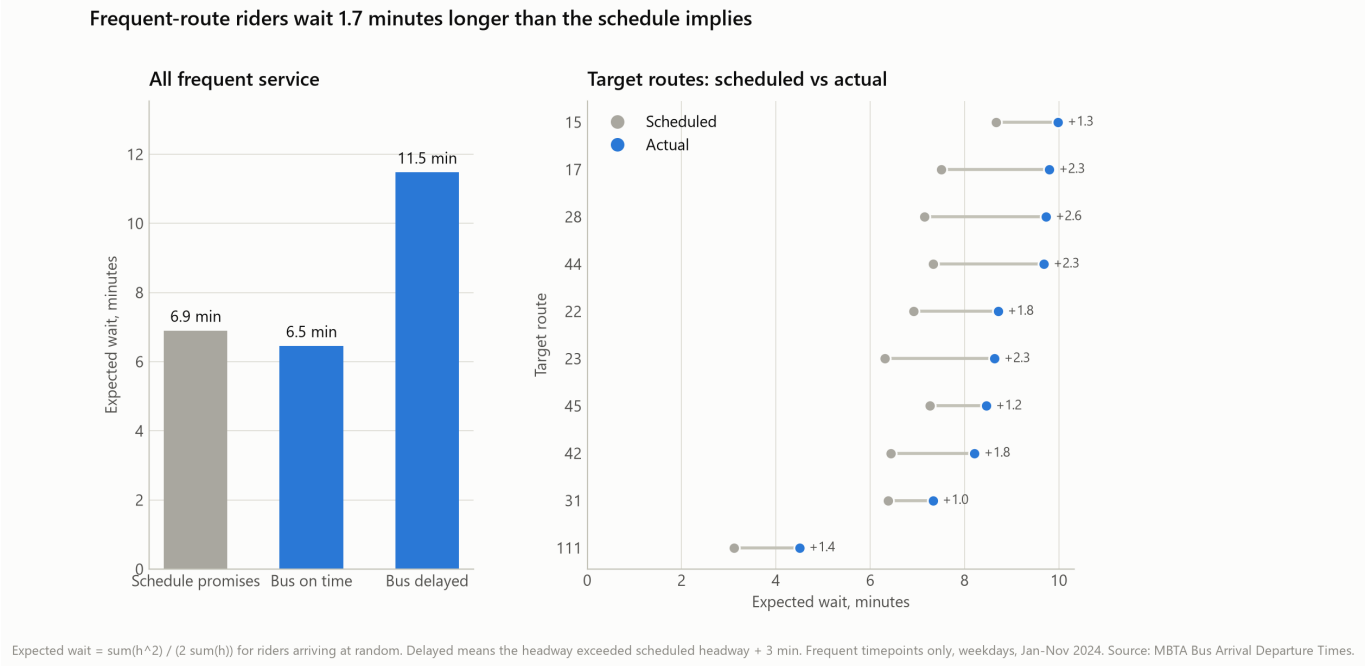
**Verdict:** Partly true: a typical trip takes about as long as scheduled, but fast and slow trips on a route differ by about 16 minutes.




Source: MBTA Bus Arrival Departure Times 2024. Trips are matched from Startpoint to Endpoint. Full table: data/processed/travel_time_all_routes.csv.

## Figure 4. Rider wait time

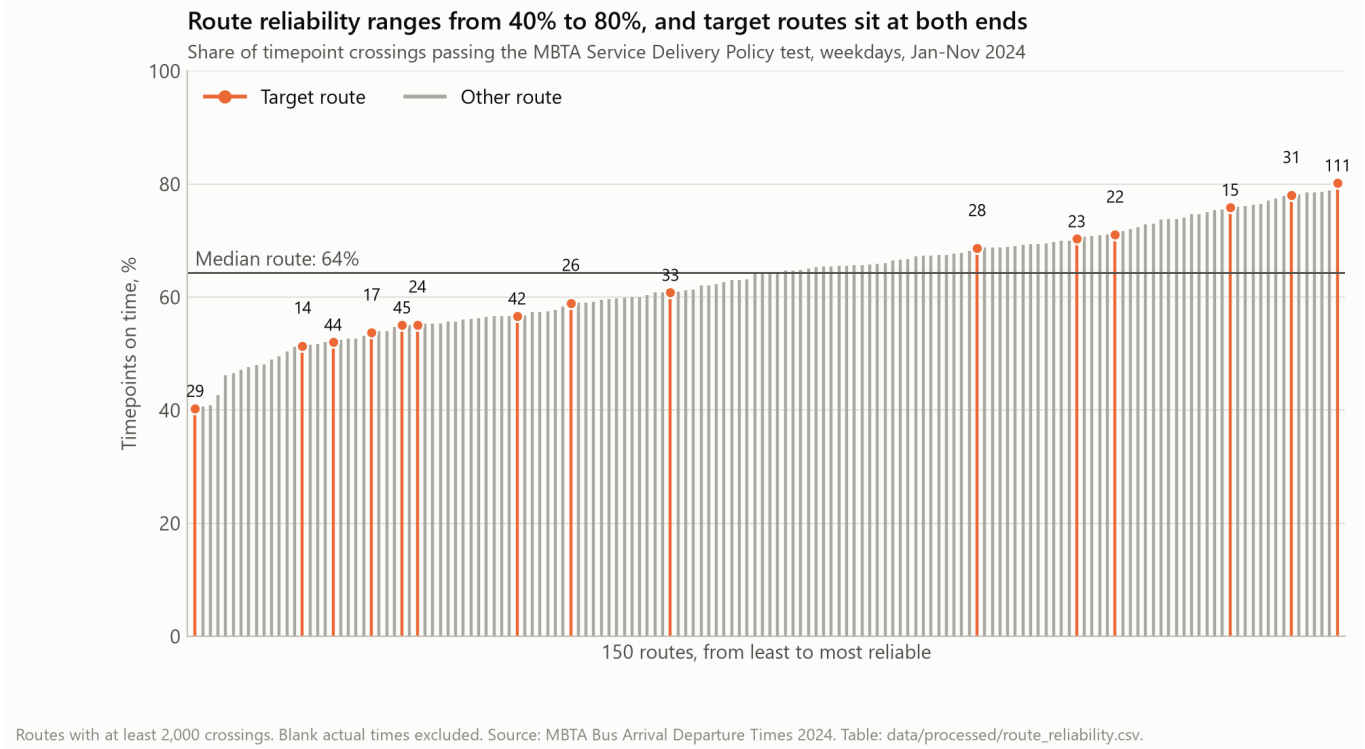
**Verdict:** True: riders on frequent routes wait about 8.6 minutes instead of the 6.9 the schedule suggests, and about 11.5 when a bus is delayed.




Expected wait = sum(h^2) / (2 sum(h)) for riders arriving at random. Delayed means the headway exceeded scheduled headway + 3 min. Frequent timepoints only, weekdays, Jan-Nov 2024. Source: MBTA Bus Arrival Departure Times.

## Figure 5. Reliability by route

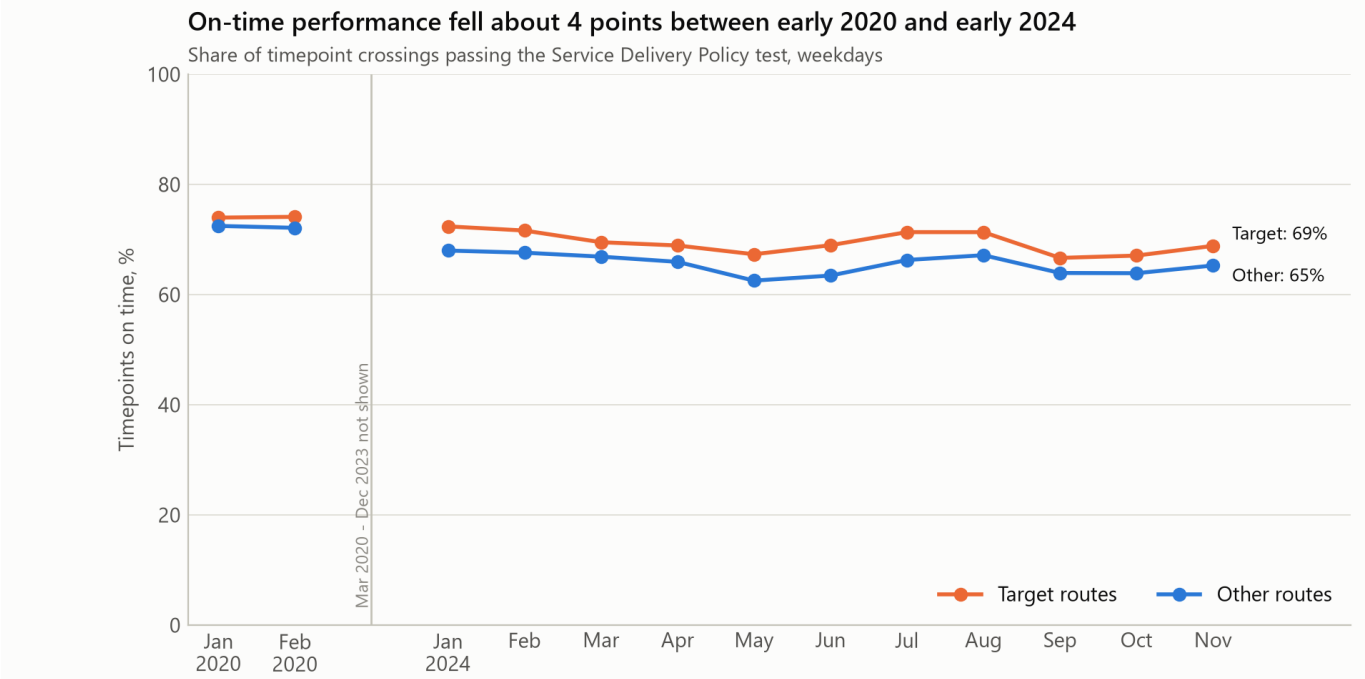
**Verdict:** True for routes overall, partly true for target routes: routes range from 40% to 80% on time, and target routes are on time about as often as others but run about 1.4 minutes later.




Routes with at least 2,000 crossings. Blank actual times excluded. Source: MBTA Bus Arrival Departure Times 2024. Table: data/processed/route_reliability.csv.

## Figure 6. Reliability over time

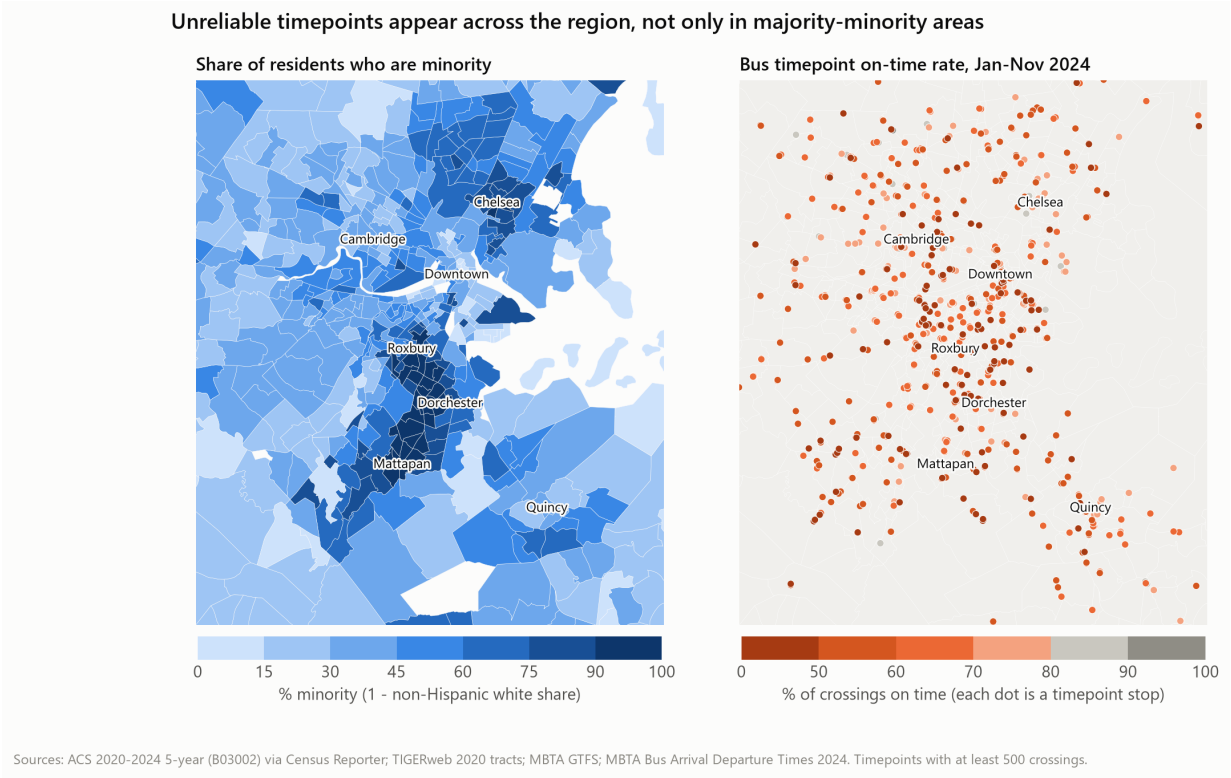
**Verdict:** True: between early 2020 and early 2024, the on-time rate fell from 73% to 69%, and average delay rose from 2.3 to 3.9 minutes.




Pooled by timepoint crossing. Source: MBTA Bus Arrival Departure Times 2020 and 2024. 2018-2019 are not publicly downloadable.

## Figure 7. Reliability and neighborhoods

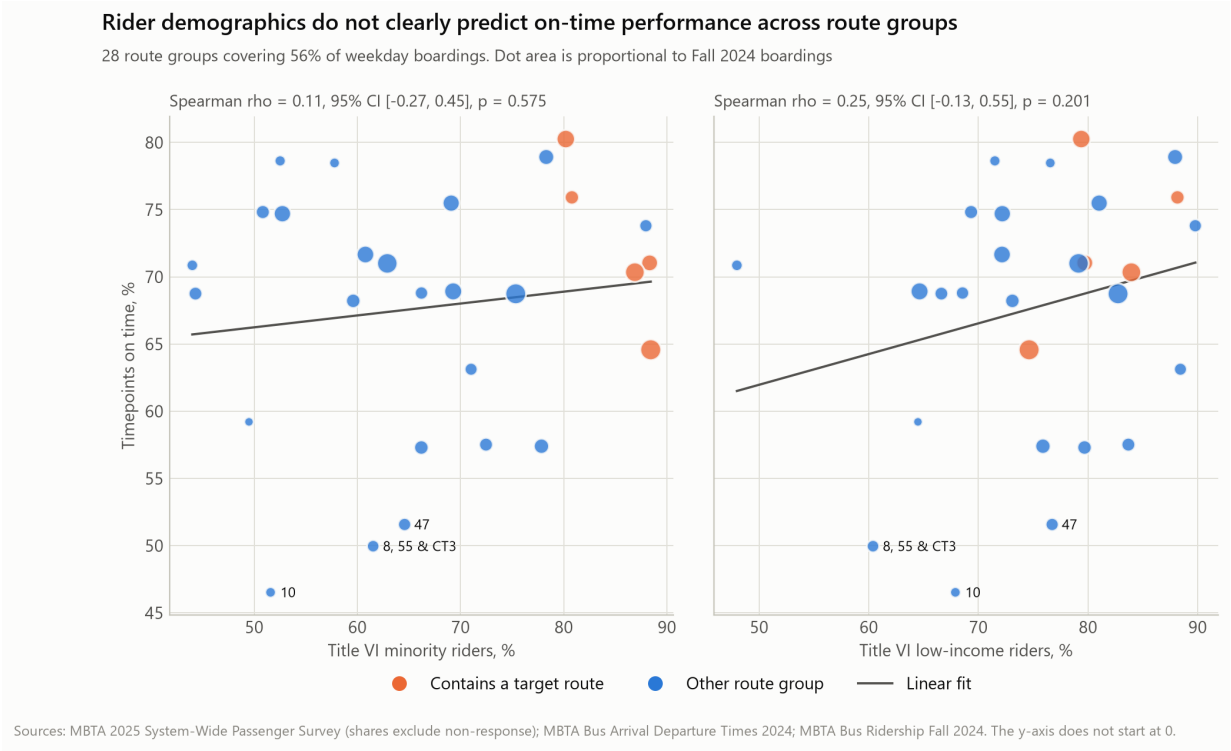
**Verdict:** No clear pattern: stops with poor on-time rates appear across the whole region, not only in mostly minority neighborhoods.




Sources: ACS 2020-2024 5-year (B03002) via Census Reporter; TIGERweb 2020 tracts; MBTA GTFS; MBTA Bus Arrival Departure Times 2024. Timepoints with at least 500 crossings.

## Figure 8. Rider demographics by route

**Verdict:** No clear link: routes with more minority or low-income riders are not clearly less on time, but this check covers only 56% of riders.



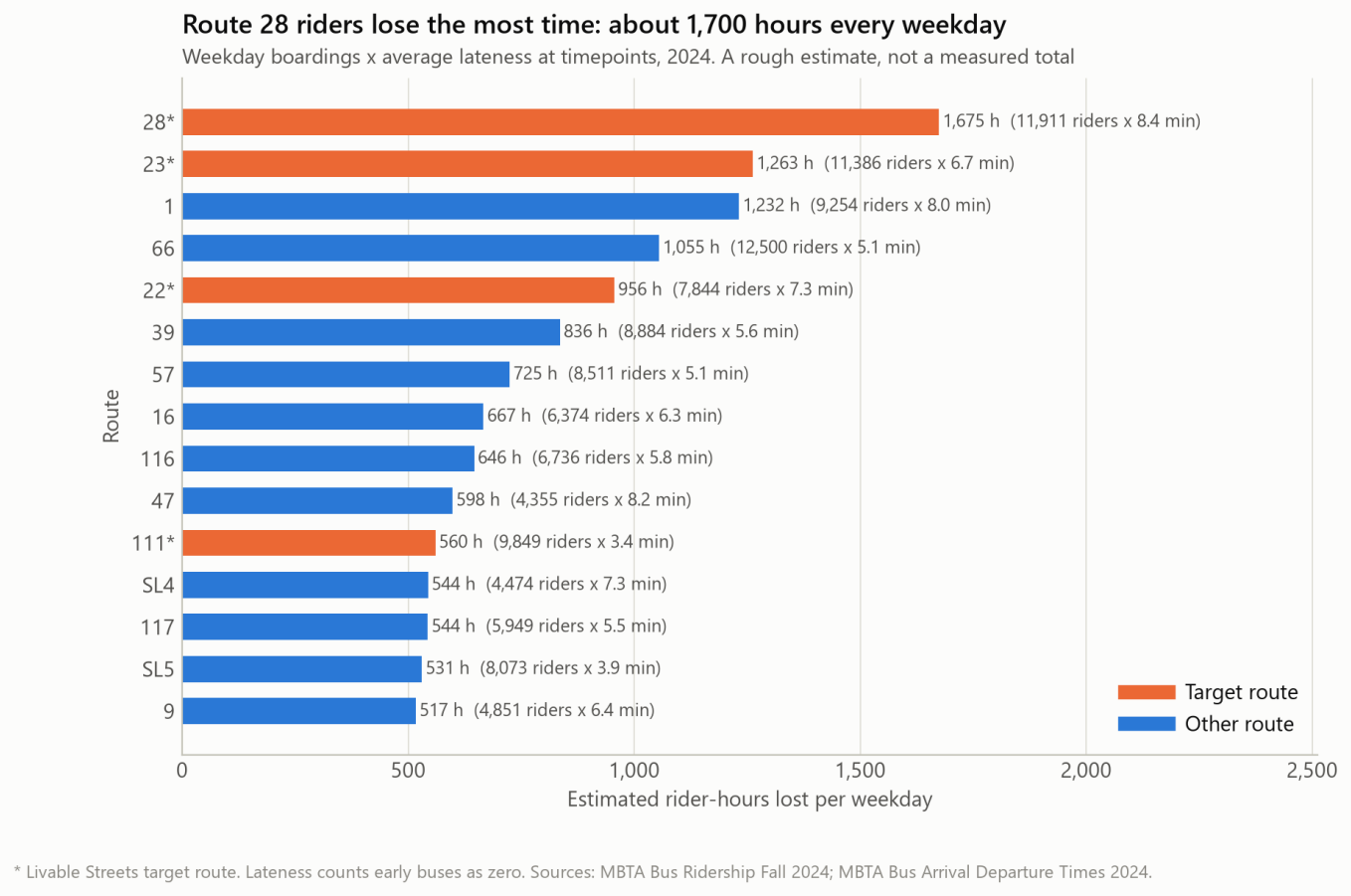

## Figure 9. Neighborhood demographics by tract

**Verdict:** Partly true: on-time rates are about the same across neighborhoods, but buses run somewhat later in minority and lower-income areas.



## Figure 10. Rider-hours lost

**Verdict:** Weakly true: the 15 routes losing the most time account for 45% of lost hours, but they also carry 38% of riders.




* Livable Streets target route. Lateness counts early buses as zero. Sources: MBTA Bus Ridership Fall 2024; MBTA Bus Arrival Departure Times 2024.